

# Love of Variety, Habit Formation, and Dynamic Advertising

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# Roadmap

- 1 Motivation
- 2 Model
- 3 Simulation Design
- 4 Results: Consumer Behavior
- 5 Next Steps

# Motivation: Data Privacy, Ad Targeting, and Consumer Agency

- Concerns around data privacy and ad targeting
  - Facebook's 5B FTC settlement
  - Google ad tracking opt-out
- Amazon anti-trust scrutiny
  - Amazon accounts for almost 40% of US e-commerce sales (Statista, 2023)
  - Uses consumer data to target ads and build market share as both retailer and platform
- Consumer agency over data (opt out)
  - Apple iOS 14.5 update: 85% opt-out rate upon rollout (Flurry Analytics, 2022)
  - Facebook ad revenue down \$13bn in 2022 due to app tracking transparency (ATT) policy

# Addressing the Concern

- How can we test the impact of policy interventions surrounding data tracking and ad targeting in a context like Amazon, where the firm has strong incentives to strategically target ads and promotions?
- Structural demand estimation is a powerful tool for policy analysis, but requires a model of consumer behavior and firm incentives

## Key Challenge

How do we model consumer behavior in a way that captures the key drivers of consumption dynamics and firms' response?

# Why Does Variety Matter?

- Canonical logit models ignore consumer preferences for variety, which are empirically important
- These preferences may shape consumer switching behavior, which in turn shapes firms' incentives to strategically target ads and promotions (i.e. to encourage switch to a higher markup product)
- Understanding the role of variety-seeking is valuable for modeling consumer behavior and evaluating the impact data privacy policies on both consumers and firms

## Research Question

How much do consumer tastes for novelty and habit influence consumption dynamics and firms' advertising strategies in a dynamic market?

# What This Paper Does

- ① **Model:** Derive a discrete-choice demand system where the utility of a product depends on both deviation from the consumer's choice history  $\theta_{it}$  and product quality  $X_{jt}$
- ② **Simulation:** Characterize consumer and firm behavior across preference regimes  $(\beta_i, \gamma_i)$  and advertising specifications
- ③ **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
- ④ **Counterfactuals (coming):** Study advertising targeting policy and welfare implications

**Today's talk:** Model + simulation results

# Literature Connections

## Demand side

- McFadden (1974): MNL foundation
- Berry, Levinsohn, Pakes (1995): RC logit
- Bronnenberg, Dubé, Joo (2021): Preference learning and experience-based priors

## Advertising / firm side

- Nerlove & Arrow (1962): Goodwill
- Erdem & Keane (1996): Learning dynamics
- Goldfarb & Tucker (2011): Targeted digital ads

## Contribution

I develop a dynamic demand model which micro-founds variety-seeking and use it to study how consumer preferences for novelty and habit shape consumption dynamics and firms' optimization strategies in a dynamic market.

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# Consumer's Problem

Consumer  $i$  chooses product  $j$  in period  $t$  to maximize:

$$U_{ijt} = \underbrace{\beta_{ij} X_{jt}}_{\text{characteristic component}} + \underbrace{\gamma_i \log(1 + \Sigma_{ijt}^2)}_{\text{novelty component}} + \underbrace{\kappa a_{jt}}_{\text{advertising}} - \underbrace{\alpha p_{jt}}_{\text{price}} + \xi_j + \varepsilon_{ijt}$$

- where  $\Sigma_{ijt} = X_{jt} - \theta_{it}$  is the deviation of product  $j$ 's characteristic from consumer  $i$ 's previously chosen characteristics  $\theta_{it}$  in period  $t$ .
- at the moment,  $\theta_{it} = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$ , or the mean of previously chosen product characteristics.

## Key parameters:

- $\beta_i > 0$ : marginal utility of quality
- $\gamma_i > 0$ : variety-seeking
- $\kappa > 0$ : advertising shifter

## Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from  $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$

# State Variable: History of Choices

## History of Choices $\theta_{it}$

$$\theta_{it} = f(\{X_{js}\}_{s < t}) = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$$

The consumer's history, for simplicity, is summarized by the mean of previously chosen product characteristics. Updates each period as the consumer makes a new choice. This is the key state variable linking past behavior to current utility.

## Initialization strategy:

- 1 **Initial state:** Simulate  $T_{\text{prior}} = 5$  pre-sample periods from  $J \in [1, 2, 3, 4, 5]$  to generate an experience-based prior. Similar to Bronnenberg, Dubé, Joo (2021).

*Initial history yields more heterogeneous behavior — more on this shortly.*

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# Simulation Parameters

**Table:** Parameters

| Param.       | Value | Description           |
|--------------|-------|-----------------------|
| $T$          | 100   | Periods               |
| $J$          | 5     | Products              |
| $S$          | 1,000 | Simulations           |
| $\kappa$     | 0     | Advertising (off)     |
| $\alpha$     | 0     | Price (off)           |
| $\xi_j$      | 0     | Quality (off)         |
| $\beta_{df}$ | 0     | Discount factor (off) |

## Preference for Characteristic $j$ :

- $\beta_i = 2$  — fixed quality sensitivity

## Preference for Variety:

- $\gamma_i \in \{6, 9, 12\}$  — low, medium, high variety-seeking

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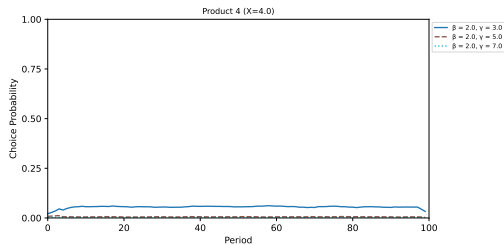
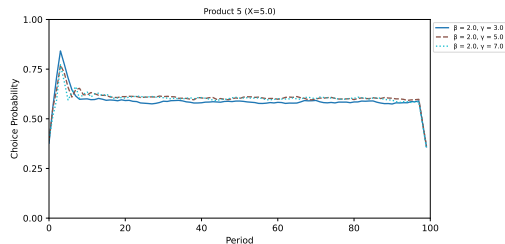
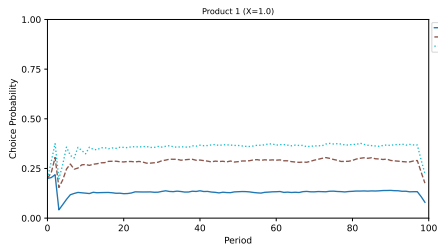
# Naive Measurement

- Near static choice probabilities
- Any variation due to  $\varepsilon_{ijt}$ , not consumer preferences
- CCPs are not informative about consumer behavior or firm incentives
- Nearly all mass on Product 5, which has the highest quality  $X_{jt}$

Table: Naive CCPs

| Product   | CCP   |
|-----------|-------|
| Product 1 | 0.02% |
| Product 2 | 0.54% |
| Product 3 | 4.05% |
| Product 4 | 22.3% |
| Product 5 | 72.0% |

# LOV Specification



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# Road to Estimation

## ① Multiple consumers

Extend simulation to  $I = 5$  consumers who have heterogeneous  $(\beta_i, \gamma_i)$ ; aggregate to market shares

## ② Consumer value function

Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values

## ③ Nielsen scanner data

Match simulated moments to data; identify  $(\beta_i, \gamma_i)$  using BLP instruments

## ④ Firm's Problem

Ad targeting policy; profit maximization; counterfactuals

# Summary

- ① **Model:** Consumer utility depends on deviation from experienced mean; nests variety-seeking and habit formation
- ② **History:** Rich behavioral dynamics; regime differences generate cross-sectional sorting
- ③ **Next:** Multiple consumers, structural estimation, Nielsen data

**Thank you — questions welcome**

## **Appendix**

# References

-  Berry, Levinsohn, Pakes (1995). "Automobile Prices in Market Equilibrium." *Econometrica*.
-  Bronnenberg, Dubé, Joo (2021). "Experience-Based Learning and the Dynamics of Brand Shares." *Journal of Political Economy*.
-  Nerlove & Arrow (1962). "Optimal Advertising Policy under Dynamic Conditions." *Economica*.
-  Erdem & Keane (1996). "Decision-Making under Uncertainty: Capturing Dynamic Brand Choice Processes in Turbulent Consumer Goods Markets." *Marketing Science*.
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# Simulation Algorithm (Consumer)

- ➊ Draw  $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$  for all  $i, j, t$
- ➋ Initialize  $\bar{X}_{i0}$  (set mean or prior mean)
- ➌ For  $t = 1, \dots, T$ :
  - ➊ Compute  $U_{ijt} = \beta_i X_{jt} - \gamma_i (X_{jt} - \bar{X}_{it})^2 + \kappa a_{jt} + \varepsilon_{ijt}$
  - ➋ Set  $j^* = \arg \max_j U_{ijt}$
  - ➌ Update  $\bar{X}_{it+1} = \frac{1}{t} \sum_{k \leq t} x_{ijk^*}$
- ➍ Compute CCPs via log-sum-exp for numerical stability:  
 $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
- ➎ Average over  $S = 1,000$  simulation draws